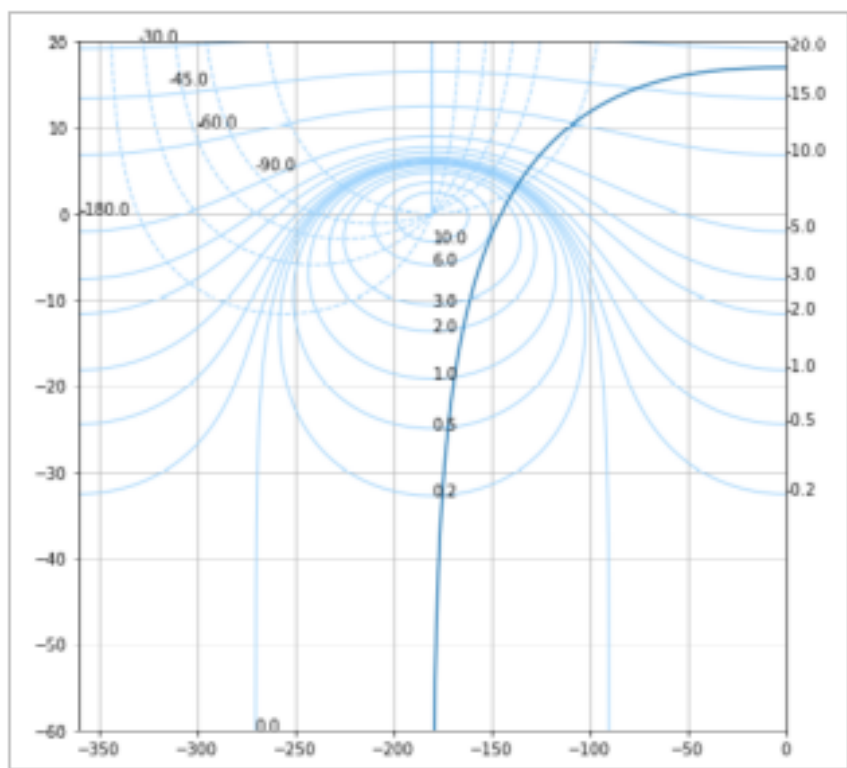
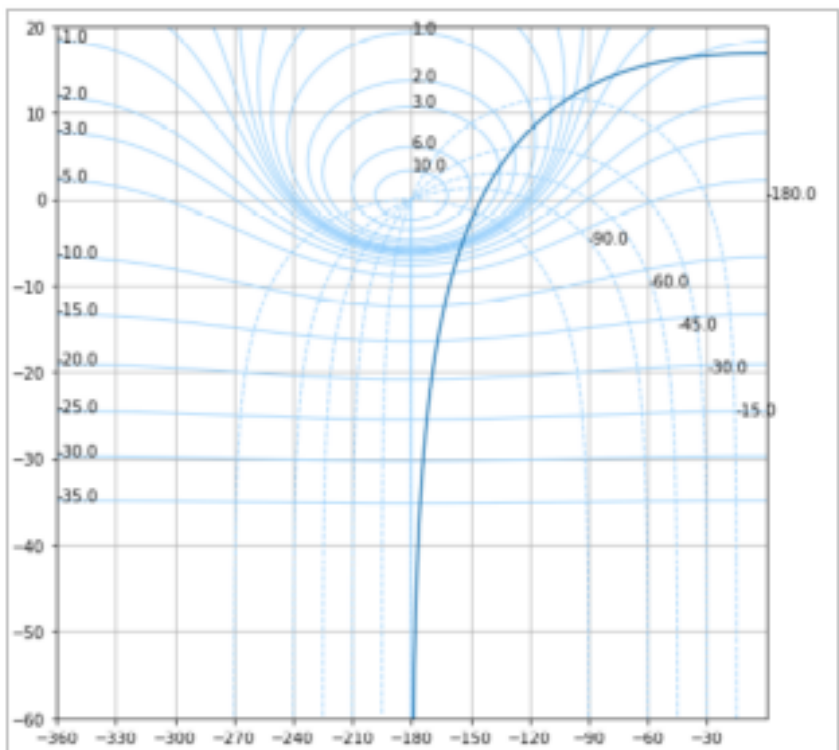


University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problem Set 5.3: Frequency Domain Analysis

Question 1

Consider the following Nichols and Inverse Nichols charts for a second-order system:

- A) Is the system stable? Give rough gain and phase margins, and explain whether you think the system's stability is robust.
- B) Roughly sketch the shape of the magnitude Bode plot for the closed-loop system. Based on your sketch, describe the damping characteristics you expect the closed-loop system's response to have, and whether it has complex poles.
- C) Another engineer warns you that the plant's behaviour is susceptible to temperature-related fluctuations. Do you expect that the closed-loop system will respond consistently despite this problem? What about high-frequency noise?
- D) Do you agree that the system would be more robust if a smaller gain was used? Would your answer change if you were told that it is to be used exclusively for very low frequency operation?



Solution:

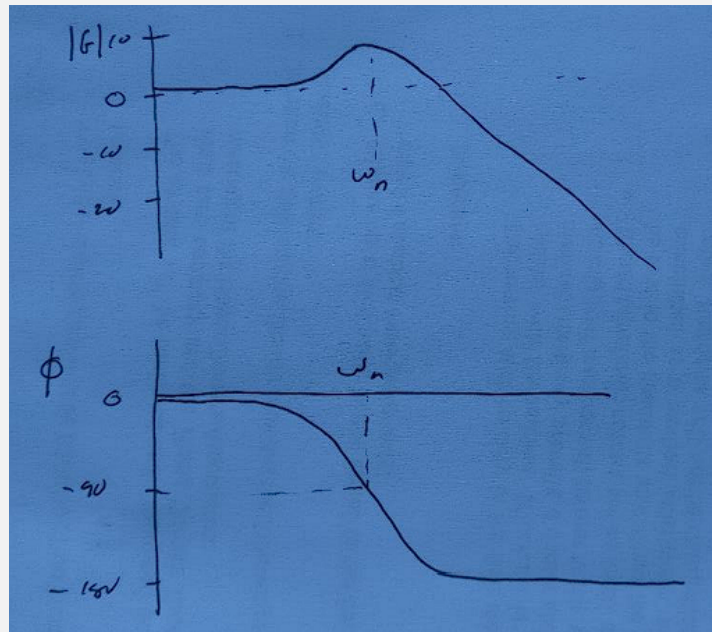
A) The system is stable.

Gain margin: 60dB

Phase margin: approx. 35 degrees

The system passes through the gain circles and gets close to the 6dB circle. It will be more sensitive at lower frequencies due to this gain.

B) The system will be underdamped due to the rise in gain around ω_n .



C) No, temperature fluctuations generally have a lower frequency and may have a significant effect. High frequency noise significantly greater than ω_n will be suppressed.

D) Yes, if a smaller gain is used the curve will be moved down. This will increase the gain margin and the stability margin of the system.

Question 2

A system has a type number greater than zero. With reference to its gain at low frequencies (< 1 rad/s) explain why it might be easier to analyse using Bode or Nichols methods, rather than a Nyquist plot.

Solution:

The Nyquist plot is generated using the magnitude and phase vector at continuously varying frequencies. As the denominator tends to zero in a system with a type number greater than zero the magnitude will tend to infinity. The vector could become too large to plot and retain the detail near the critical point.

A Nichols or Bode plot would be preferable because the log-based scale would reduce the range of the output at low frequency and still carry important stability information at the same time.

Question 3

Use a program of your choice to construct Bode, Nyquist, Nichols and Inverse Nichols plots for the following systems:

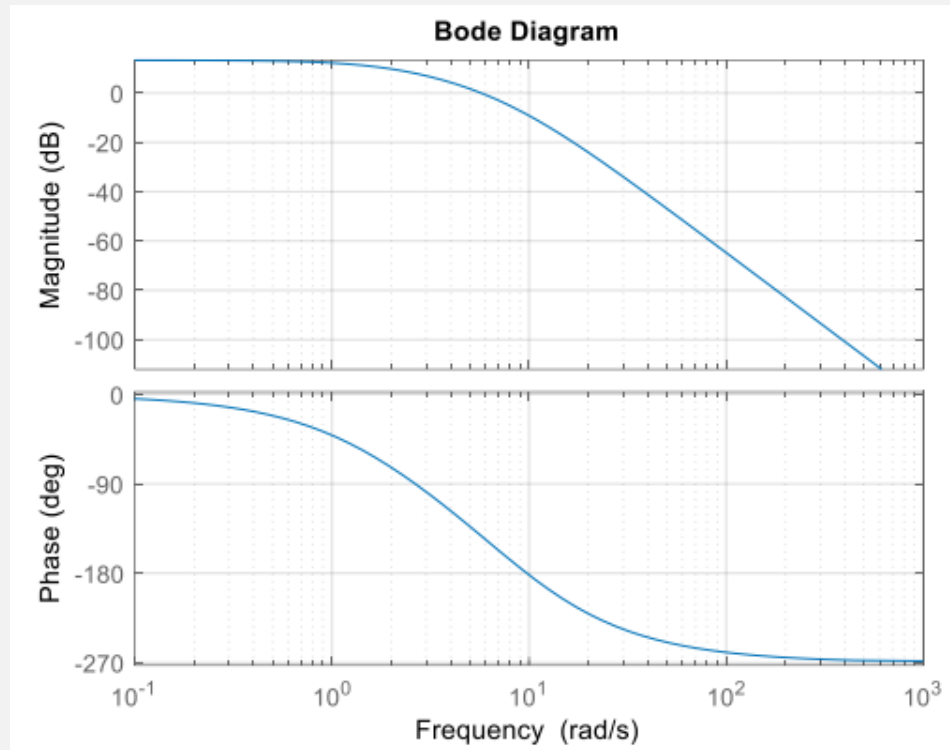
- i) $\frac{582.9}{s^3+18s^2+95s+126}$
- ii) $\frac{39.27s^2+392.7s+1335}{s^3+20s^2+141s+360}$
- iii) $\frac{229.6s-229.6}{s^4+11s^3+46s^2+86s+60}$

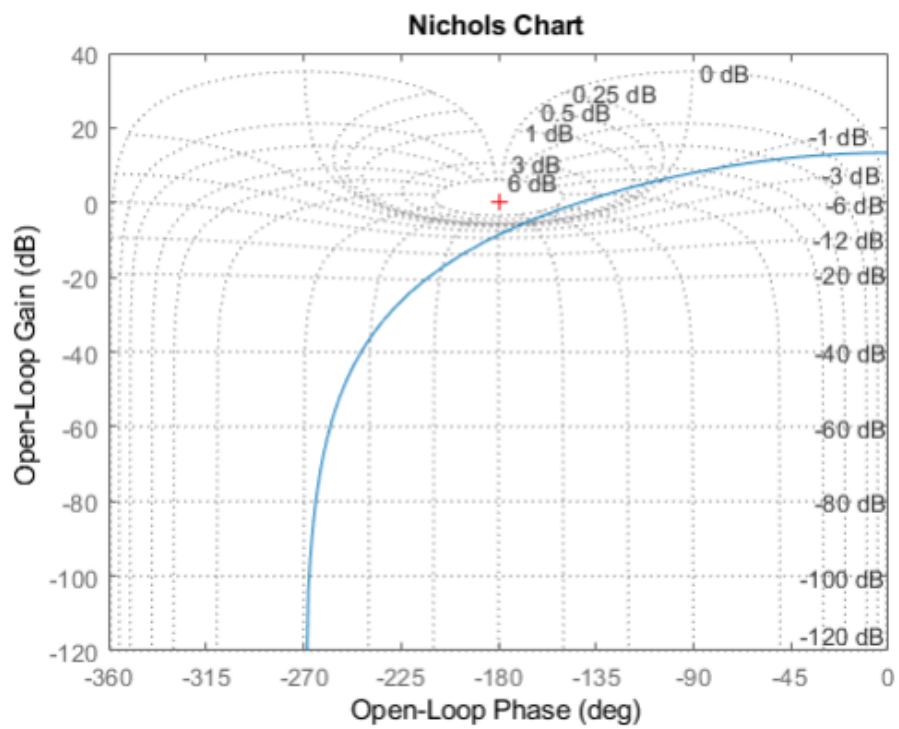
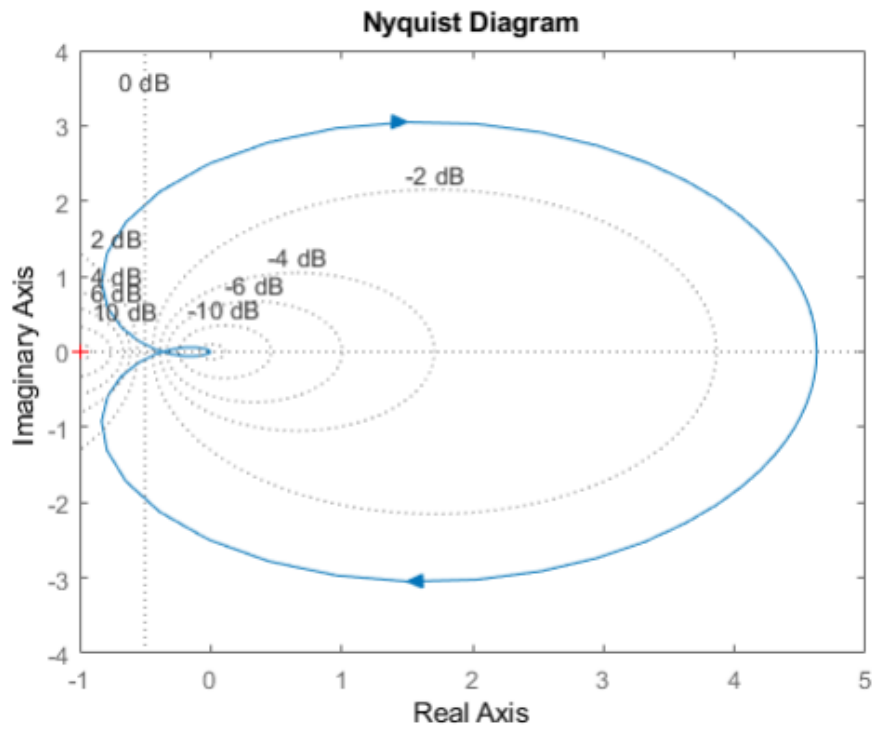
Based on your plots, write a short assessment of the system considering the following characteristics:

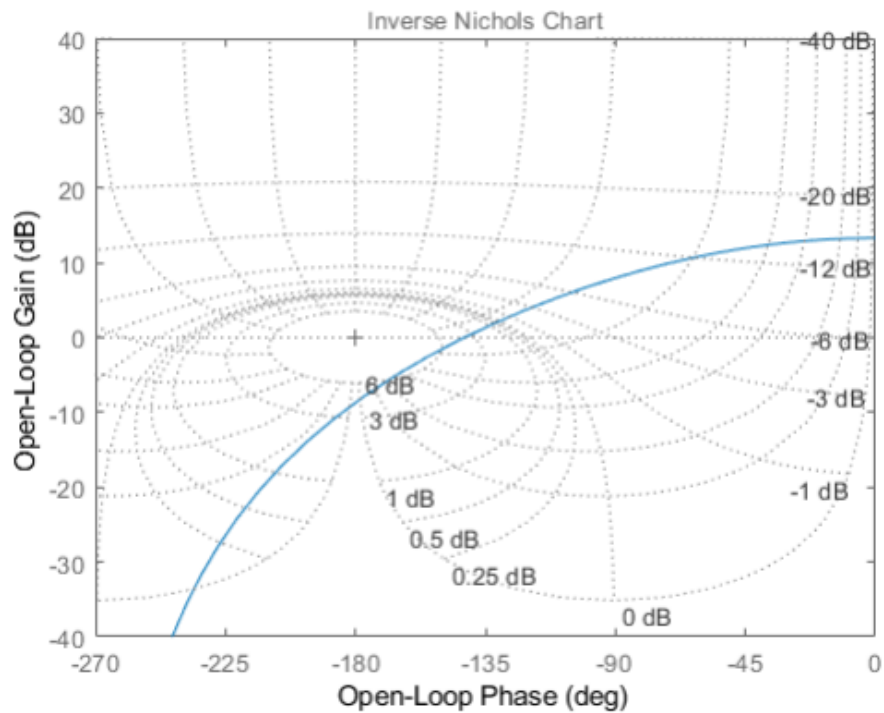
- A) **Stability.** If it is stable, identify its stability margins, and explain whether you perceive them to be an accurate reflection of how stable the system is. If it is not, evaluate whether it can be stabilized by changing the gain.
- B) **Frequency response:** describe whether the system amplifies or attenuates low frequencies, high frequencies and any other frequency ranges you find interesting. Is the system vulnerable to high-frequency noise? If the system is unstable and can be stabilized, you may consider the stabilized version.
- C) Repeat (B), but for the closed-loop response.
- D) **Robustness:** are gain or phase fluctuations more dangerous for the system? Do you expect the system's behaviour to remain consistent if the open-loop model's parameters fluctuate? Again, you may consider a stabilized model if necessary.

Solution:

System i)







A) The system is stable but not robustly.

- Gain margin: 10 dB
- Phase margin: 45 degrees

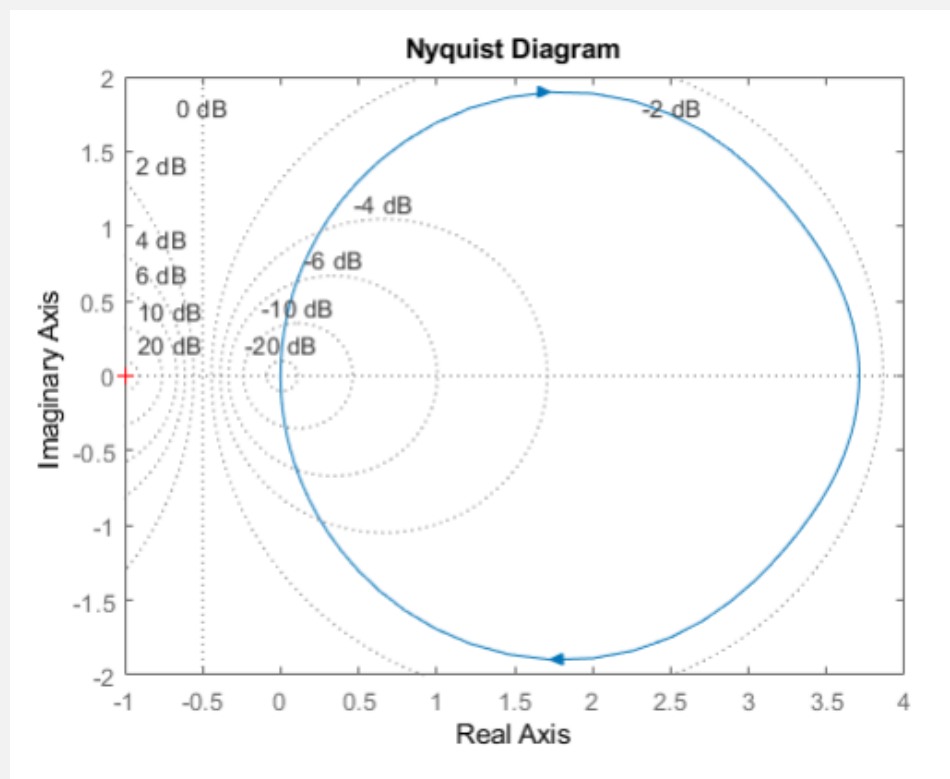
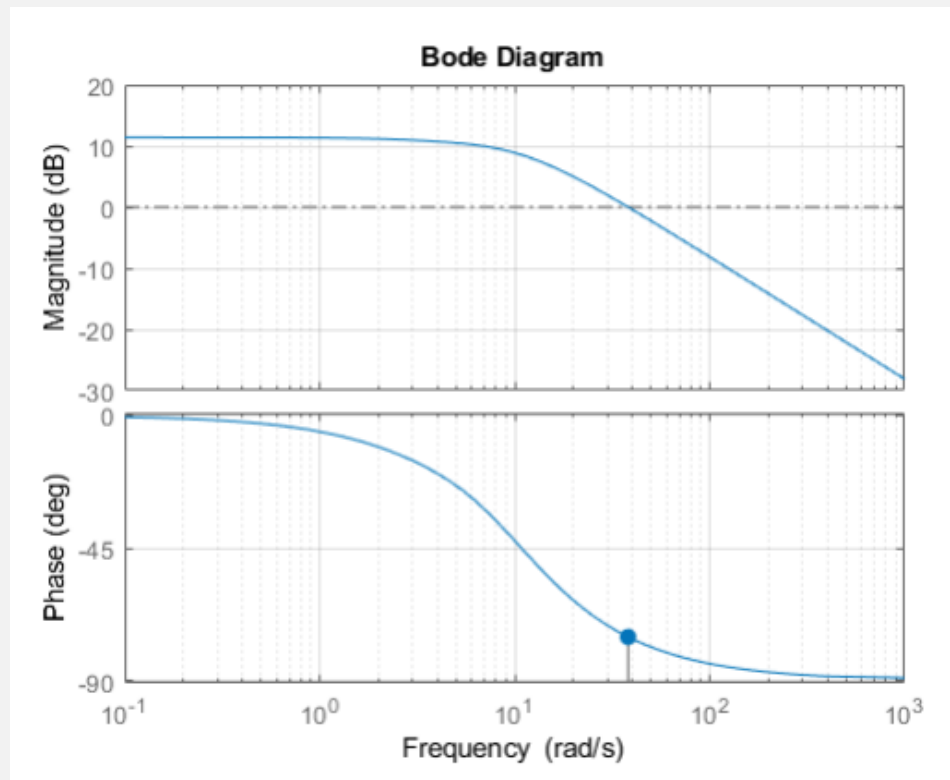
These margins don't accurately represent the stability. The Nichols chart shows that the system is particularly sensitive to gain and phase between 6 and 9 rad/s.

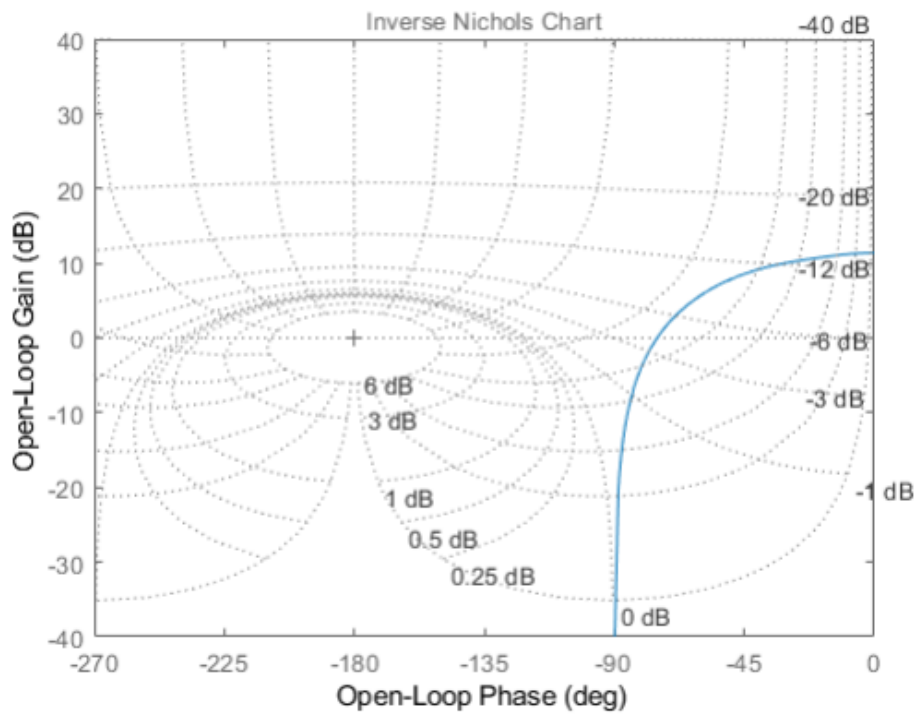
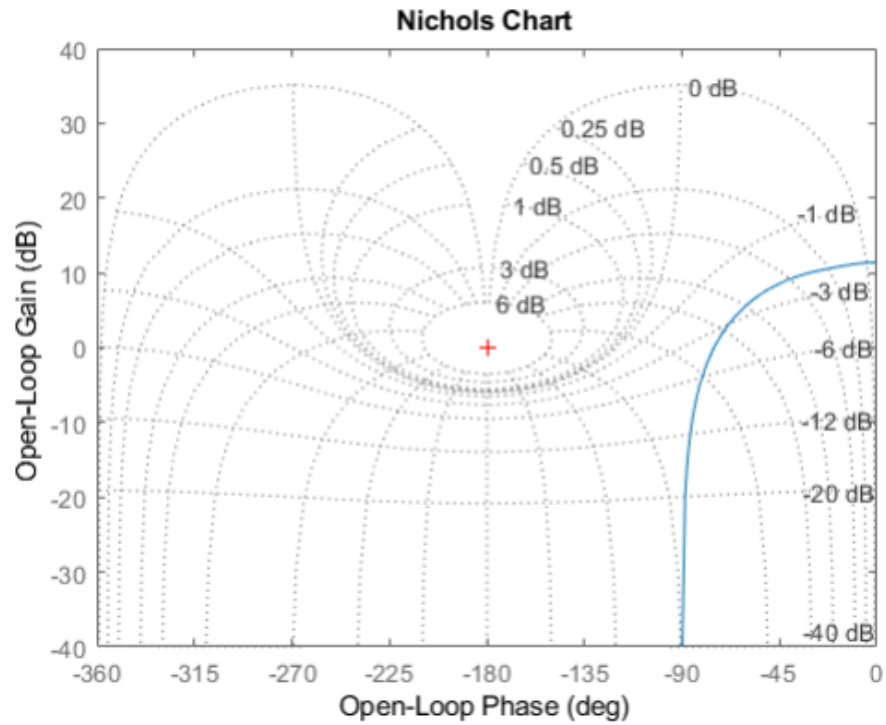
B) The system amplifies low frequencies and attenuates high frequencies. It won't be particularly susceptible to HF noise.

C) In closed loop most frequencies are attenuated apart from 6 to 9 rad/s where gain is applied. Noise in this region could result in instability.

D) Both gain and phase disturbances could cause instability. Note how the inverse Nichols chart cuts through the 6 dB circle. A small disturbance in gain or phase could push the system to instability.

System ii)





A) The system is stable. The Nyquist and Nichols plots show that clearly.

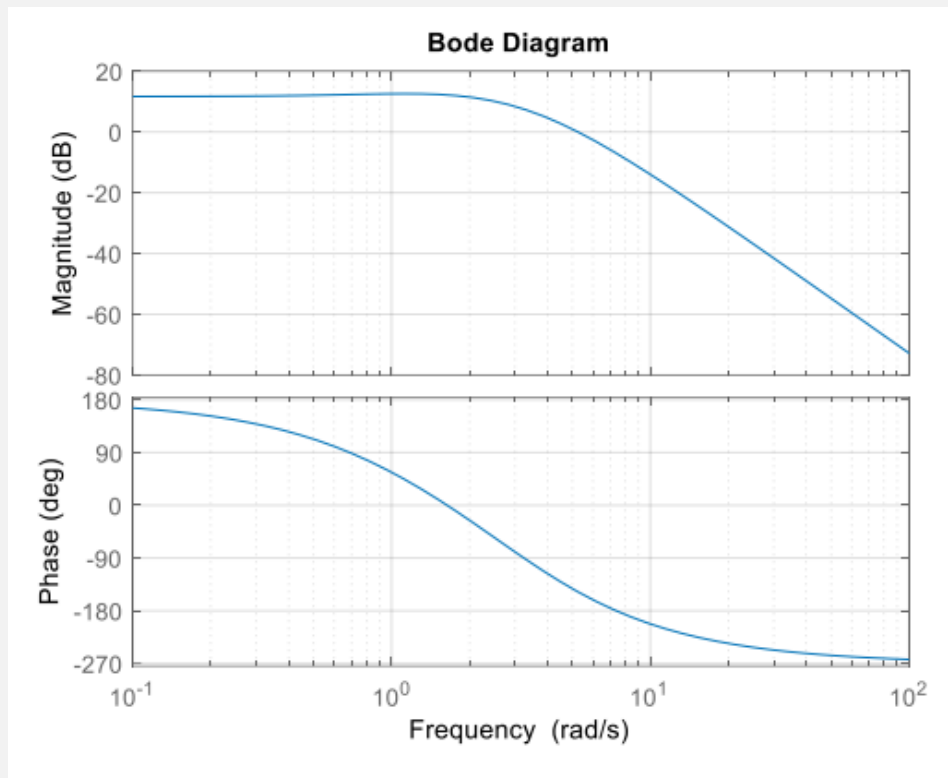
- Phase margin: 100 degrees
- Gain margin: Infinite

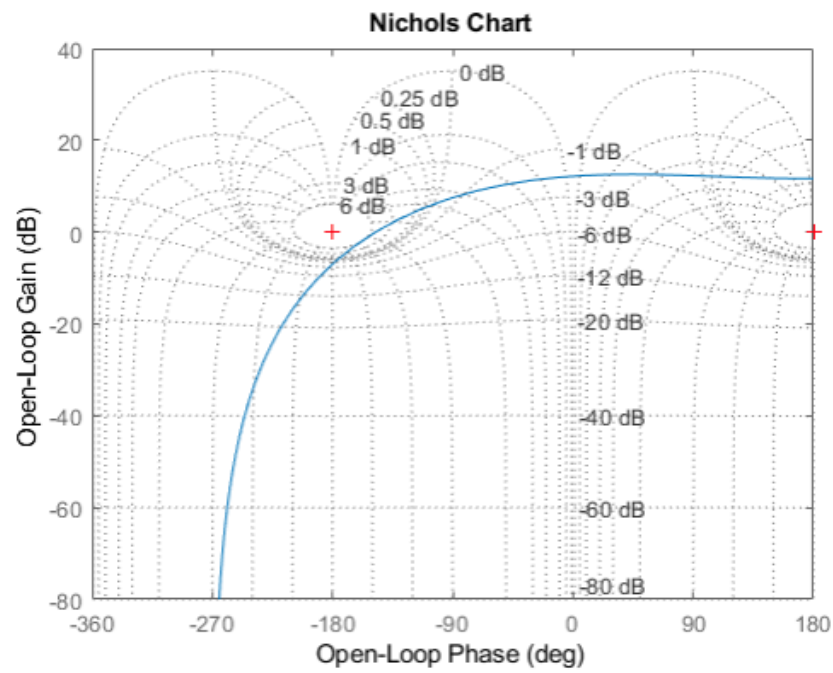
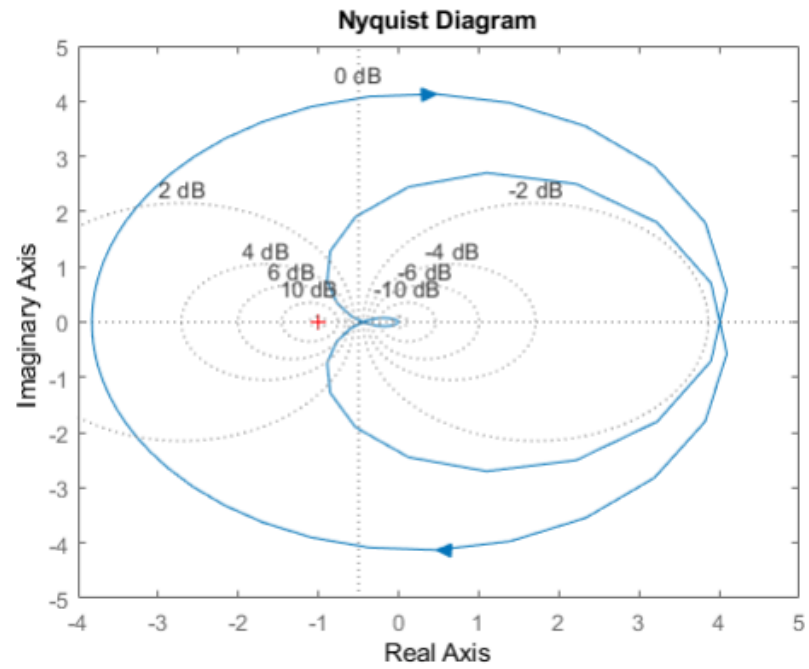
B) The system amplifies low frequencies and attenuates high frequencies. There are no frequencies that are particularly critical to the stability of the system.

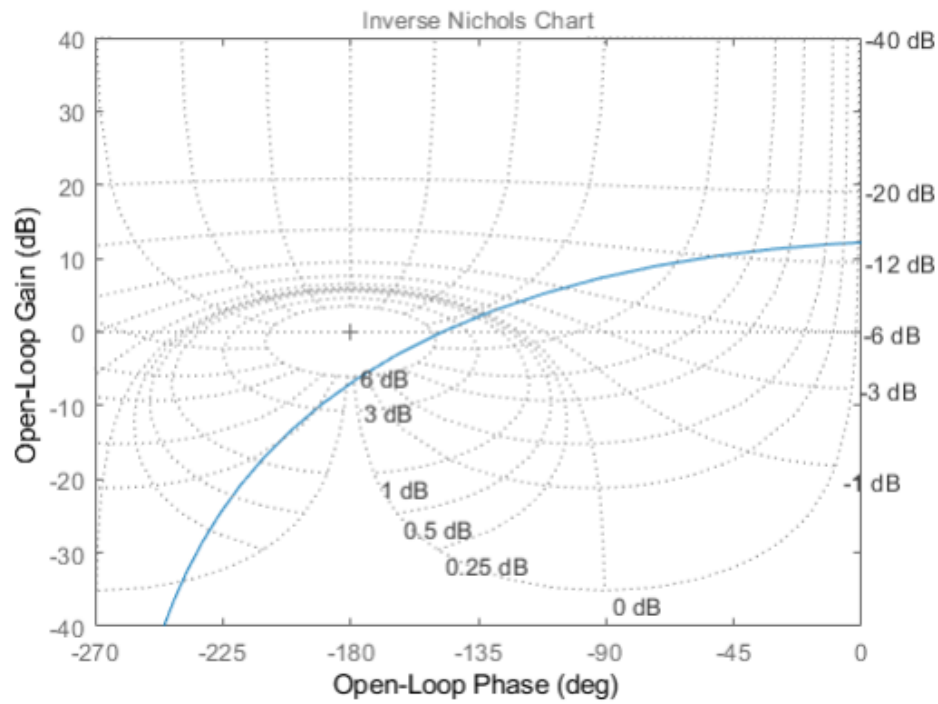
C) In the closed loop response, all frequencies are attenuated with higher frequencies being suppressed more than lower frequencies. Again, no frequencies are particularly critical to stability.

D) The system is robust to gain fluctuations. A significant phase fluctuation could introduce instability.

System iii)







A) The system is unstable. The Nyquist plot loops around -1. Looking at the Nichols chart if the gain is reduced the curve will shift down. This will increase the stability margins of the system and could stabilize it. The Nyquist plot almost reaches -4. Therefore, a gain reduction of at least 0.25 must be applied to stabilize it. At this point the system has a large phase margin of about 180 degrees and a gain margin of 18 dB.

B) The system attenuates high frequencies and passes low frequencies with little change.

C) In closed loop higher frequencies are attenuated more than low frequencies. The system is particularly sensitive to gain or phase disturbances between 2 and 6 rad/s.

D) The system is sensitive to both gain and phase changes, especially at frequencies near 4 rad/s. The robustness can be improved by reducing the gain further.